PROJECT 2

# Vulnerability Assessment Report

*Use TryHackMe to scan a target and document your findings like a real security analyst*

| Your Name: | Date: | TryHackMe Username: |
|---|---|---|
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**What is this project?**

Security analysts run vulnerability scans to find weaknesses in systems before attackers do.
In this project, you will use TryHackMe , a free, browser-based platform , to complete a guided hacking/scanning lab. No software to install. No coding required. Just a web browser!
When done, you will write up your findings in this report template.

## Part 1: Set Up TryHackMe (Do This First)

Follow these steps exactly. Take your time , there is no rush.

**1**

### Create your free account

Go to: https://tryhackme.com
Click 'Join Free' and create an account (email or Google sign-in)
You do NOT need to pay for anything , the free tier is enough for this project

**2**

### Find the right room

Once logged in, click the search bar at the top
Search for: 'Vulnversity' (this is the recommended room for beginners)
Alternatively, search for: 'Basic Pentesting' or 'Blue'
Click the room and click 'Join Room'

**3**

### Start the machine

Inside the room, click the green 'Start Machine' button
Wait 2-3 minutes for the machine to boot
TryHackMe will give you a target IP address (like 10.10.x.x) - write it down!
You will use this IP address throughout the lab

| 4 | **Use the in-browser attack machine**<br>Click 'Start AttackBox' at the top of the page<br>This opens a full Linux computer inside your browser - no installation needed!<br>All the tools you need are already installed on this machine |
|---|---|

**Having trouble? That is completely normal and okay!**

TryHackMe has a built-in hint system - click the hint button on each task if you get stuck
The TryHackMe Discord community is very friendly and beginner-focused
Ask your instructor - getting stuck is part of learning, not a sign of failure
Important: You do NOT need to complete every task in the room. The goal is to practice scanning and document what you find. Even partial completion is valuable and counts for this project!

## Part 2: Run Your Scan and Take Notes

As you work through the TryHackMe room, answer these questions. Write notes as you go.

### 2A. Initial Reconnaissance

| Target IP Address (given by TryHackMe) |
|---|
| *10.67.150.195* |

| What Nmap command did you run? (or what scan did the room guide you to do?) |
|---|
| *nmap -sV 10.67.150.195* |

| What open ports did you find? List all of them. |
|---|
| *Ports 21 (FTP), 22 (SSH), 139 (NetBIOS-SSN), 445 (NetBIOS-SSN), 3128 (HTTP Proxy), and 3333 (HTTP)* |

### 2B. Identify the Top 3 Findings

From your scan results, pick the 3 most interesting or risky findings to document below.
*Do your best, you cannot get this wrong. There are no perfect answers here.*

## FINDING #1

| **Port / Service:** | **Severity (circle one):** |
|---|---|
| 22 (SSH) | LOW MEDIUM HIGH CRITICAL |

**What is this vulnerability? Explain it in plain English:**

*This port allows communication between a user and a server and is more secure than its counterpart, Telnet. Most of the vulnerability with SSH comes from the user, such as weak credentials or poor management of access.*

*Source:*

*SSH — Port 22 (TCP/UDP) | WhatPortIs*

**Why is this a problem? What could an attacker do with this?**

*If SSH is exploited, an attacker could gain access to a company's remote servers, allowing them unrestricted access to company data. This level of access can be used in Man-in-the-Middle attacks; if the attacker is able to gain root access, attacks like rootkits are possible.*

*Source:*

*SSH Hack: Malware & Hackers Collect SSH Keys to Spread Attack*

**Recommended Fix:**

*Strong passwords, monitor access, and change default port*

## FINDING #2

| **Port / Service:** | **Severity (circle one):** |
|---|---|
| 3128 (HTTP-Proxy) | LOW MEDIUM HIGH CRITICAL |

**What is this vulnerability? Explain it in plain English:**

*This port caches website data for future use, allowing for faster access later when the website is accessed again by a user. By using this port, bandwidth and response times are improved, since the server doesn't need to travel as far to fetch the data for a website.*

*Source:*

*Squid Proxy HTTP — Port 3128 (TCP) | WhatPortIs*

**Why is this a problem? What could an attacker do with this?**

*This port uses HTTP rather than HTTPS, meaning data is stored in plain text. Attackers are able to read the plaintext, making the data being transmitted extremely vulnerable. Attackers may also use cache poisoning to change the data of a website, making the site look legitimate while being extremely dangerous. Any data input into this fake site would be compromised by the attacker, which might give them access to other systems within a company.*

**Recommended Fix:**
*Enable encryption, keep systems patched, and monitor access*

**FINDING #3**

| **Port / Service:** | **Severity (circle one):** |
|---|---|
| 3333 (HTTP) | LOW MEDIUM HIGH CRITICAL |

**What is this vulnerability? Explain it in plain English:**
*This port is used to capture and display caller identification information, such as phone number, caller name, and the time stamp of the phone call. This captured information is displayed in real time to clients, making it easier to monitor activity.*

*Source:*
*Network Caller ID — Port 3333 (TCP) | WhatPortIs*
**Why is this a problem? What could an attacker do with this?**
*Like 3128, this port is unencrypted, meaning the data captured is in plaintext and easy to read. An attacker could intercept this data, eavesdrop on conversations, or use this system as a means of deploying a DoS attack.*

**Recommended Fix:**
*Patching, enable encryption via SSH or VPN, network segmentation*

## Part 3: Executive Summary

Imagine you are presenting your findings to a manager who does not know tech. Write a short summary.

*Use plain language. Pretend the reader is a smart person who just does not know security terms.*

**Overall Risk Level of This System (circle one and explain why)**

*LOW RISK MEDIUM RISK HIGH RISK CRITICAL RISK*
*Explain: While each of these issues may be relatively low risks on their own, an attacker would only need to successfully hack one of these ports to have high levels of access to the company's data. For example, an attacker may gain access to the company's web server, allowing them to log all*

*passwords that employees enter. Using these passwords, the attacker is able to impersonate an employee and pretend to be them during phone calls, giving them the ability to use social engineering tactics to receive even more information that is harmful to the company.*

**The 3 most important things this organization should fix, in order of priority:**

1. *Ensure all data is encrypted*
2. *Change ports to non-default ones*
3. *Employee training on cybersecurity*

**Write 3-5 sentences summarizing what you found (imagine you are the analyst presenting this to leadership):**

*During this vulnerability assessment, I identified three areas of weakness in the company's infrastructure that leaves both employee and customer data at a high risk of attack. The web server used by the company, utilizing the SSH protocol, is particularly vulnerable due to the potential for weak passwords or unmonitored access to company resources. The web server also uses an unencrypted proxy, making it extremely easy for an attacker to poison the cache that employees use to access company websites. Finally, the company's VoIP system is also unencrypted, leaving any calls vulnerable to eavesdropping or unauthorized monitoring. To reduce the risk of attack, the following steps should be implemented: ensure all company data is encrypted, change all utilized ports from their default configurations, and ensure all employees receive training on cybersecurity concepts.*

## Part 4: Reflection

| What was the hardest part of this lab? | What Security+ topic did this lab connect to most? |
|---|---|
| *Of the assigned sections of this lab, tasks 1 and 2 on TryHackMe, I didn't struggle with anything outside of not being familiar with the website. However, I attempted the other tasks and was not able to complete tasks 4 or 5. To complete task 3, I was able to make it most of the way through, and was able to guess the last question correctly based on the other answers provided.* | *Vulnerability management and port scanning* |

# Before You Submit: Checklist

### Check each item when complete:

- [ ] I created a TryHackMe account and joined a room
- [ ] I recorded my target IP address
- [ ] I documented at least 3 findings (port, severity, explanation, fix)
- [ ] I wrote an executive summary in plain language
- [ ] I completed the reflection section

Bonus: Take a screenshot of your TryHackMe progress bar and attach it!
Portfolio tip: Save as PDF. Title it: 'Vulnerability Assessment Report - [Your Name] - [Date]'
Add your TryHackMe profile link to your LinkedIn: https://tryhackme.com/p/[yourusername]

